

41891: Cloud Computing Infrastructure
42891: Infrastructure for Cloud Computing

Subject Information

Lecturer and Coordinator:
Dr. Haimin Zhang

Teaching Staffs

- * Coordinator and Lecturer:

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- * Tutors:

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Overview

- * Lecture – 1 hour per week
 - * Different topics are introduced (Week 1-6, and Week 8, 9, 10)
 - * Two Quizzes (Week 7 and Week 11)
- * Tutorials/Labs – 2 hours per week (starting from Week 2)
 - * Lab exercises + possible tutorial questions
- * Self study/assignment – 7 hour per week
 - * Lectures + labs are NOT enough!

Assessments (1)

- * Assessment Task 1:

- * Weight: 20%

- * Assignment 1: Individual

- * An individual written assignment based on lecture content

- * Released in Week 2

- * Due: **Friday of Week 5**

- * Late submissions:

- * A late penalty of 20% per working day will apply.

- * Special consideration for late submission must be arranged by Week 4. Formal documentation must be provided.

Assessments (2)

- * Assessment Task 2:

- * Weight: 40%

- * Assignment 2: Group major project

- * Students are required to specify, design, architect, develop and implement a major infrastructure based project.

- * More analysis, comparison, and discussion based on the understanding of existing cloud computing techniques are expected from **Postgraduates**.

- * Start forming groups in Week 3 (four students in a group)

- * Released in Week 6

- * Due: Week 11 (implementation 25%, report 15%)

- * Late submissions:

- * A late penalty of 20% per working day will apply.

- * Special consideration for late submission must be arranged by Week 9. Formal documentation must be provided.

Assessments (3)

- * Assessment Task 3:

- * Weight: 20%

- * Quiz 1

- * An open-book individual assessment.

- * Multiple-choice questions and possibly short-answer questions.

- (theory exam; questions will be based on the key concepts presented in lectures/labs)

- * Due: Week 7

Assessments (3)

- * Assessment Task 3:

- * Weight: 20%

- * Quiz 2

- * An open-book individual assessment.

- * Multiple-choice questions and possibly short-answer questions.

- (theory exam; questions will be based on the key concepts presented in lectures/labs)

- * **Postgraduates** need to answer an essay question.

- (may be from the recommended texts and/or reference materials)

- * Due: Week 11

Assessments (5)

To pass this subject,
Students **MUST** to get **50% or more** of the total marks!

Overview of lectures

- * Week 1: Cloud Infrastructure: An Introduction
- * Week 2: Cloud Virtualisation Technologies
- * Week 3: Cloud Architecture and Management Mechanisms
- * Week 4: Cloud Virtualisation and NaaS
- * Week 5: Data Centre Fundamentals
- * Week 6: Management of Data Centre
- * Week 7: Quiz 1
- * Week 8: Security and Compliance in the Cloud
- * Week 9: Public/Private Cloud Comparison; and Review
- * Week 10: Public holiday, No lecture
- * Week 11: Quiz 2
- * Week 12: Project presentations and evaluations, No lecture

NOTE: The weekly schedule may be adjusted during the semester to suit the needs of the class

Overview of labs

- * Lab 1 (Week 2): get familiar with the environment: Build VMs, Vmware ESXI
- * Lab 2 (Week 3): iSCSI and NFS on ESXI
- * Lab 3 (Week 4): vCenter Server and VM migration (Form project groups)
- * Lab 4 (Week 5): vSwitches
- * Lab 5 (Week 6): Resources (1)
- * Lab 6 (Week 7): Work on learning contract for Assignment 2, Discuss with your tutor
- * StuVac
- * Lab 7 (Week 8): Confirm learning contract with your tutor
- * Lab 8 (Week 9): Discuss with your tutor about Assignment 2 solution
- * Lab 9 (Week 10): Discuss with your tutor about Assignment 2 solution
- * Lab 10 & 11 (Week 11&12): Project presentations and evaluations

NOTE: The weekly schedule may be adjusted during the semester to suit the needs of the class

Overview of labs

Other cost:

- * When using the FEIT computer lab for this subject, you are recommended to bring:
 - * USB external hard disks with at least 64GB free to run/save their work. A USB 3.0 SSD (solid state drive) is recommended.